

✓ AI in Apps

Applications Observed

App	How AI is Used
Google Maps	Predicts traffic and best routes
YouTube	Recommends videos
Google Assistant	Understands voice commands

Real-Life Understanding

Google Maps

AI analyzes traffic, distance, and road conditions to suggest the fastest route.

YouTube

AI observes watched videos and recommends similar content using machine learning.

Google Assistant

AI converts voice into text, understands commands, and gives smart responses.

Basic AI Concepts

Machine Learning

AI learns from data and improves automatically.

Example:

- YouTube learns what videos users watch
 - Then recommends similar videos
-

Data Input

AI takes different types of input such as:

- Voice
- Text
- Photos
- Location

Example: Saying "Navigate home" to Google Assistant.

Prediction

AI predicts what users may want.

Examples:

- YouTube suggesting videos
- Instagram showing reels you like

Conditional Logic in AI

AI uses conditions to make decisions.

Example Python code:

```
if user_likes_music:
    recommend_music_video
else:
    recommend_general_video
```

Meaning:

- If the condition is true, AI shows related content.

Real Example:

- YouTube recommendations

```
user_likes_music = True

if user_likes_music:
    print("Recommend music videos")
else:
    print("Recommend general videos")
```

Recommend music videos

✓ Loops in AI

Loops repeat tasks many times automatically.

Example:

- AI checking thousands of videos quickly

Simple Python Example:

```
for i in range(5):
    print("AI Processing Data")
```

This repeats 5 times.

```
for i in range(5):
    print("AI Processing Data")
```

AI Processing Data
AI Processing Data
AI Processing Data
AI Processing Data
AI Processing Data

✓ Arrays / Data Storage in AI

AI stores user preferences in lists or arrays.

Example:

- Favorite movies
- Liked videos
- Search history

Python Example:

```
movies = ["Action", "Comedy", "Sci-Fi"]
```

AI uses this stored data to recommend similar content.

```
movies = ["Action", "Comedy", "Sci-Fi"]

print(movies)

['Action', 'Comedy', 'Sci-Fi']
```

✓ Voice and Text Processing in AI

Example Voice Command: "Play songs"

How AI Processes It:

1. Converts voice into text
2. Understands the meaning
3. Gives the correct output

Applications Using This:

- Google Assistant
- Siri
- Alexa

```
voice_command = "Play songs"

print("Voice Command:", voice_command)
print("AI converts voice to text and understands the command")
```

```
Voice Command: Play songs
AI converts voice to text and understands the command
```

```
# AI in Daily Life Observation Task

apps = ["Google Maps", "YouTube", "Google Assistant"]

for app in apps:
    print(app, "uses AI in daily life")
```

```
Google Maps uses AI in daily life
YouTube uses AI in daily life
Google Assistant uses AI in daily life
```

✓ AI in Daily Life – Observation Report

Applications Observed

- Google Maps
- YouTube
- Google Assistant

Observations

1. Google Maps uses AI for route prediction.
2. YouTube recommends videos using machine learning.

3. Voice assistants process voice input using AI.
4. AI uses loops and data arrays to process large information.
5. Conditional logic helps AI make smart decisions.

Conclusion

AI is used in many daily applications to make tasks easier, faster, and smarter.

```
import numpy as np
import pandas as pd

print("AI Environment Ready")
```

AI Environment Ready